

Halo communication in the mEVP sea-ice solver of the FESOM2 GPU port

N. Koldunov, 6 August 2026 — summary of measurements

Three ways of exchanging the halo less often

mEVP advances the ice momentum in 120 sub-cycles per model time step, and the ice-velocity halo is at present exchanged after each of them. We implemented and measured three alternatives.

Delayed exchange — exchange every K -th sub-cycle and accept a halo up to $K - 1$ sub-cycles old. This is the largest saving of the three, but it leaves the sub-domain boundaries visible in the ice field. *Wide halo* (Danilov) — exchange every K -th sub-cycle, and recompute a K -deep ring of ghost cells in between so that no value is stale; the result reproduces the every-sub-cycle trajectory bit for bit. *Divergence form* (Danilov) — carry $\nabla \cdot \sigma$ at nodes rather than σ at elements; it agrees with $\nabla \cdot \sigma$ to 10^{-15} relative, and its effect on run time differs between the two backends.

Cost, measured against the model step and against the ice

Each number is given twice. Sea ice is 5 to 13% of a CPU model step, so an ice-only change is divided by about ten before it reaches the step, and the share of the ice cost (ice, ice dynamics and ice advection, computation together with time spent waiting for communication) is the measure appropriate to a sea-ice scheme.

		GPU		CPU, at scale	
		ice cost	model step	ice cost	model step
delayed exchange	$K=4$	−37 to −58%	−8.5 to −14%	−5 to −25%	−0.3 to −3.6%
wide halo, exact	$K=8$	−12 to −19%	−2 to −7%	never profitable	
divergence form alone	—	+2 to +17%	0 to +1.6%	−2 to −8.5%	−0.5 to 0%

On CPU the saving grows with core count (−5% of the ice at 128 cores, −25% at 2048); on GPU it falls as each GPU holds more of the mesh (−58% at 32 000 surface nodes per GPU, −37% at 460 000). The divergence form removes 2 to 8.5% of the ice cost from 512 cores upward, and is invisible in the model step only because of the denominator. On GPU it costs, because the sub-cycle there is limited by how often work is dispatched to the device rather than by arithmetic or memory traffic, so removing element arrays removes work that was not being paid for. Danilov’s $\sim 20\%$ for the divergence form with its double halo corresponds to our −19.1% of the ice cost for the same combination (fArc, 16 GPU nodes).

Imprint of the domain decomposition

Under the delayed exchange the stale values lie on the one-cell boundary of each sub-domain, and the decomposition is fixed for the run, so the same nodes are perturbed at every step. Measured on fArc, 16 GPUs, 60 days with daily output: mean $|\Delta \text{ ice thickness}|$ on the boundary and in the interior.

	boundary	interior	ratio
two identical runs (control)	0.00105 m	0.00124 m	0.85
wide halo, exact, $K=8$	0.00104	0.00120	0.87
delayed exchange, $K=2$	0.00432	0.00249	1.70
delayed exchange, $K=4$	0.01178	0.00452	2.52

The difference is concentrated on the boundary from the first day (ratio 3.1), remains so for two months, and appears on maps as a network of differences following the sub-domain edges. Halving the delay to $K=2$ reduces the ratio but does not remove it. The two exact configurations, which differ from the reference only in the order of summation, sit together at 0.85 and 0.87; that they do is what makes the measurement interpretable. An artificial polygonal structure in the ice field, in the shape of the domain decomposition, is not acceptable in a sea-ice model. Figure 4 of the full note gives the profiles and a map.

The integral diagnostics do not show it. Over a year on CORE2 the ice concentration and thickness correlate with the reference at 0.999995 and 0.999994, within the spread between two independent implementations of mEVP, and hemispheric ice extent departs by 0.23% of its seasonal amplitude. A feature one cell wide along sub-domain edges covers a small fraction of the mesh and leaves a domain-wide correlation near 0.99999. The accuracy protocol we had prepared would not have detected it.

Recommendation

For production we would use the exact wide halo. It is bit-for-bit identical to the standard scheme, so it requires neither an accuracy argument nor the check above, and it removes 12 to 19% of the ice cost on GPU at $K=8$. On CPU it is not profitable on any mesh we tried.

We do not regard the delayed exchange as usable, although it saves about three times as much: it places the domain decomposition in the ice field.

The divergence form is worth having on CPU at large core counts and is a cost on GPU. It is also what makes the wide halo cheapest to carry, since everything sub-cycled is then a node quantity.